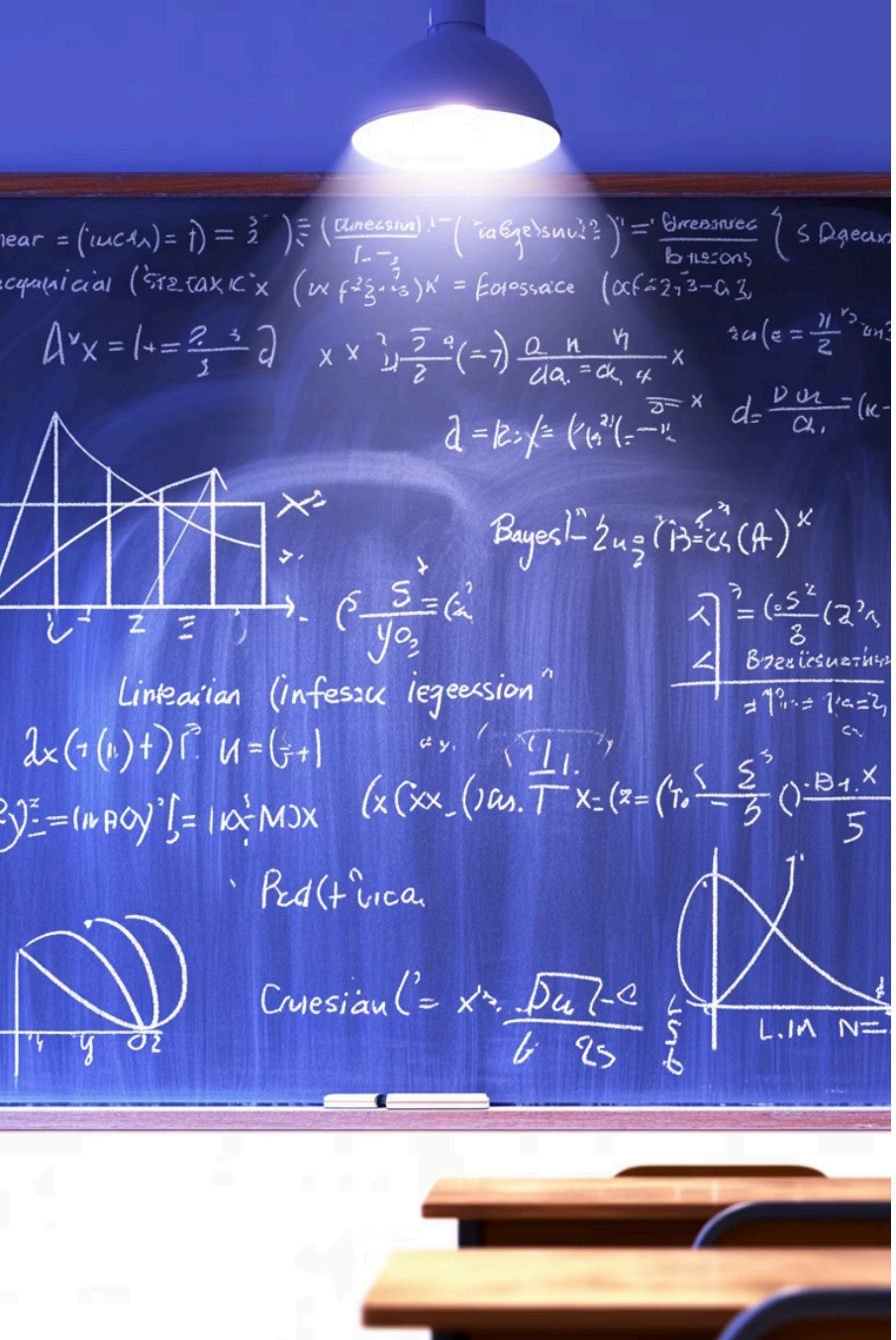


Mean, Variance, and Standard Deviation

Understanding the fundamental measures of central tendency and variability in statistical analysis

Population vs Sample: The Foundation

Population

The **entire group** of subjects we're interested in studying. For example, if we're researching polar bears in the Arctic, the population would be every single polar bear in that region.

Denoted by capital **N** for the number of subjects.

Sample

A **subset** of the population. Since studying every polar bear would be nearly impossible, we might select only 25 bears and use this smaller group to draw conclusions about the entire population.

Denoted by lowercase **n** for the number of subjects.

Understanding whether you're working with a population or sample is crucial, as we use different formulas for variance and standard deviation depending on which one we have.

Mean: The Centre of Our Data

Population Mean (μ)

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

When we have data from the **entire population**, we use capital N and the Greek letter mu (μ) to represent the mean.

Sample Mean (x̄)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

When working with a **sample**, we use lowercase n and x-bar (x̄) to represent the mean. This is pronounced "x-bar".

Notice how the formulas are identical in structure, but we use different notation to distinguish between population and sample calculations.

Variance: Measuring Spread from the Mean

Variance measures **how far the data is spread** from the mean. Think of it as capturing how much individual data points deviate from the central value.

01

Calculate deviations

Find how far each point is from the mean: $(x_i - \mu)$ for population or $(x_i - \bar{x})$ for sample

02

Square the deviations

Square each deviation to eliminate negative values and emphasise larger differences

03

Find the average

Sum all squared deviations and divide by the appropriate denominator

Population vs Sample Variance Formulas

Population Variance (σ^2)

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

We divide by **N** when we have the complete population. The symbol σ^2 is pronounced "sigma squared".

Sample Variance (s^2)

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}$$

We divide by **n-1** for samples to get unbiased variance. This gives us a more accurate estimate.

Why n-1? Dividing by n tends to underestimate variance, whilst dividing by n-2 overestimates it. Using n-1 provides the most accurate estimate, known as unbiased sample variance.

Worked Example: Teachers' Ages

A school principal wants to know the average age of teachers. She samples 10 teachers and records their ages:

Data: 43, 35, 38, 56, 29, 33, 46, 63, 49, 40

01

Calculate Sample Mean

$$\bar{x} = \frac{43 + 35 + 38 + 56 + 29 + 33 + 46 + 63 + 49 + 40}{10} = \frac{432}{10} = 43.2$$

02

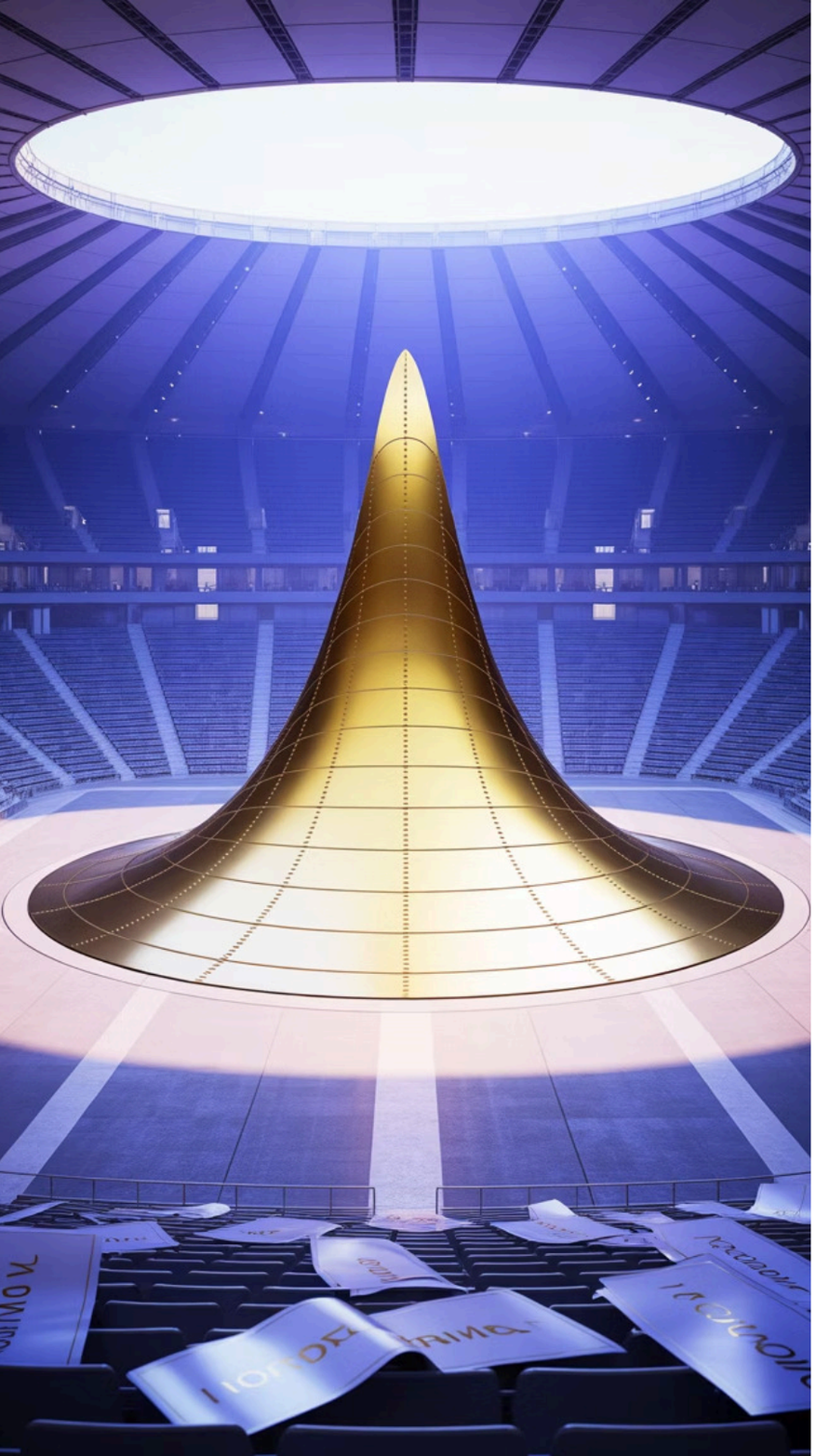
Set up deviation table

Next, we'll calculate each data point's deviation from the mean and square it

Calculating Sample Variance: Step-by-Step

Data (x_i)	Deviation (x_i - $\bar{x}$)	Squared Deviation
43	43 - 43.2 = -0.2	0.04
35	35 - 43.2 = -8.2	67.24
38	38 - 43.2 = -5.2	27.04
56	56 - 43.2 = 12.8	163.84
29	29 - 43.2 = -14.2	201.64
33	33 - 43.2 = -10.2	104.04
46	46 - 43.2 = 2.8	7.84
63	63 - 43.2 = 19.8	392.04
49	49 - 43.2 = 5.8	33.64
40	40 - 43.2 = -3.2	10.24

$$s^2 = \frac{1007.6}{10-1} = \frac{1007.6}{9} \approx 111.96$$



Standard Deviation: The Square Root of Variance

Standard deviation measures **how much data varies from the mean**. It's simply the square root of variance, bringing our measure back to the same units as our original data.

Population Standard Deviation (σ)

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

Sample Standard Deviation (s)

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Important note: Taking the square root of unbiased sample variance does reintroduce some bias, but this is generally acceptable for practical purposes.

Completing Our Example: Standard Deviation

Returning to our teachers' ages example, we found that the sample variance was approximately 111.96. Now we can calculate the standard deviation:

$$s = \sqrt{s^2} = \sqrt{111.96} \approx 10.58$$

43.2

Mean Age

Average age of teachers in the sample

111.96

Sample Variance

Squared units measuring spread

10.58

Standard Deviation

Typical deviation from the mean (in years)

The standard deviation of 10.58 years tells us that, on average, individual teachers' ages deviate from the mean by about 10.6 years.

Summary: Key Formulas and Concepts

Measure	Population	Sample
Subjects	N	n
Mean	$\mu = \Sigma x_i/N$	$\bar{x} = \Sigma x_i/n$
Variance	$\sigma^2 = \Sigma(x_i-\mu)^2/N$	$s^2 = \Sigma(x_i-\bar{x})^2/(n-1)$
Standard Deviation	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$

Remember: Use n-1 for sample variance to get unbiased estimates. Standard deviation returns our measure to the original units, making it easier to interpret than variance.

These fundamental concepts form the building blocks for more advanced statistical analysis and hypothesis testing.